

1. Deep-Dive Mathematical Derivation from First Principles

In a Cash Equities Central Risk Book (CRB) or Transaction Cost Analysis (TCA) seat at Nomura, you manage large inventory blocks across fragmentation points where execution delays directly convert into variance risk. When an Executive Director asks you to prove the Itô Isometry, they are testing if you understand why stochastic integrals are variance-preserving transformations. This is the bedrock metric for pricing execution risk, sizing liquidity premiums, and setting optimal liquidation horizons.

Here is the step-by-step mathematical proof from first principles, formulated to survive a rigorous quantitative desk review.

Step 1: Definition of the Itô Integral via Simple (Step) Processes

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, where W_t is a standard 1-Dimensional Brownian motion adapted to the filtration $\mathcal{F}_t$.

Let H_t be a bounded, adapted **simple stochastic process** (also known as a step process) over the temporal domain $[0, T]$. We define a partition Π of the interval $[0, T]$ as:

$$\Pi = \{0 = t_0 < t_1 < t_2 < \dots < t_n = T\}$$

The simple process H_t is constant on each sub-interval $[t_k, t_{k+1})$:

$$H_t = H_{t_k} \quad \text{for } t \in [t_k, t_{k+1})$$

Where H_{t_k} is explicitly $\mathcal{F}_{t_k}$ -measurable. For such a simple process, the Itô stochastic integral is defined from first principles as the finite sum of forward increments:

$$I(H) = \int_0^T H_t dW_t = \sum_{k=0}^{n-1} H_{t_k} (W_{t_{k+1}} - W_{t_k}) = \sum_{k=0}^{n-1} H_{t_k} \Delta W_k$$

Where $\Delta W_k = W_{t_{k+1}} - W_{t_k}$.

Step 2: First-Moment Properties: Expectation of the Stochastic Integral

Before calculating variance, we must verify that the expectation of the stochastic integral is zero. Taking the expectation of our finite sum definition:

$$\mathbb{E} \left[\int_0^T H_t dW_t \right] = \mathbb{E} \left[\sum_{k=0}^{n-1} H_{t_k} \Delta W_k \right]$$

By the linearity of the expectation operator:

$$\mathbb{E} \left[\sum_{k=0}^{n-1} H_{t_k} \Delta W_k \right] = \sum_{k=0}^{n-1} \mathbb{E}[H_{t_k} \Delta W_k]$$

We apply the **Law of Total Expectation** (Tower Property) by conditioning each term on the local filtration $\mathcal{F}_{t_k}$:

$$\mathbb{E}[H_{t_k} \Delta W_k] = \mathbb{E} [\mathbb{E}[H_{t_k} \Delta W_k \mid \mathcal{F}_{t_k}]]$$

Since H_{t_k} is $\mathcal{F}_{t_k}$ -measurable, it behaves as a known constant within the inner conditional expectation and can be pulled outside:

$$\mathbb{E} [\mathbb{E}[H_{t_k} \Delta W_k \mid \mathcal{F}_{t_k}]] = \mathbb{E} [H_{t_k} \mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}]]$$

By the structural properties of Brownian motion, the increment $\Delta W_k = W_{t_{k+1}} - W_{t_k}$ is completely independent of the historical filtration $\mathcal{F}_{t_k}$, and its unconditional expectation is zero:

$$\mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}] = \mathbb{E}[\Delta W_k] = 0$$

Substituting this back into our summation yields:

$$\mathbb{E}[H_{t_k} \Delta W_k] = \mathbb{E}[H_{t_k} \cdot 0] = 0 \implies \mathbb{E} \left[\int_0^T H_t dW_t \right] = 0$$

Step 3: Second-Moment Properties: The Core Isometry Proof

We now evaluate the expectation of the *squared* stochastic integral:

$$\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \mathbb{E} \left[\left(\sum_{k=0}^{n-1} H_{t_k} \Delta W_k \right)^2 \right]$$

Expanding the square of a summation using the algebraic identity $(\sum a_k)^2 = \sum a_k^2 + 2 \sum_{j < k} a_j a_k$:

$$\mathbb{E} \left[\left(\sum_{k=0}^{n-1} H_{t_k} \Delta W_k \right)^2 \right] = \mathbb{E} \left[\sum_{k=0}^{n-1} H_{t_k}^2 (\Delta W_k)^2 + 2 \sum_{0 \leq j < k \leq n-1} H_{t_j} H_{t_k} \Delta W_j \Delta W_k \right]$$

Applying linearity of expectation partitions this into diagonal components and off-diagonal cross-product components:

$$\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \sum_{k=0}^{n-1} \mathbb{E} [H_{t_k}^2 (\Delta W_k)^2] + 2 \sum_{j < k} \mathbb{E} [H_{t_j} H_{t_k} \Delta W_j \Delta W_k]$$

Step 4: Line-by-Line Dissection of Cross-Product Terms ($j < k$)

Let us isolate a single cross-product expectation term $\mathbb{E} [H_{t_j} H_{t_k} \Delta W_j \Delta W_k]$ where $j < k$.

We condition the expectation on the filtration at the start of the final interval, $\mathcal{F}_{t_k}$, using the Tower Property:

$$\mathbb{E} [H_{t_j} H_{t_k} \Delta W_j \Delta W_k] = \mathbb{E} [\mathbb{E}[H_{t_j} H_{t_k} \Delta W_j \Delta W_k \mid \mathcal{F}_{t_k}]]$$

Let us assess the measurability of each term relative to $\mathcal{F}_{t_k}$:

- H_{t_j} is $\mathcal{F}_{t_j}$ -measurable. Since $j < k$, $\mathcal{F}_{t_j} \subset \mathcal{F}_{t_k}$, so H_{t_j} is $\mathcal{F}_{t_k}$ -measurable.
- $\Delta W_j = W_{t_{j+1}} - W_{t_j}$ is $\mathcal{F}_{t_{j+1}}$ -measurable. Since $j < k \implies j+1 \leq k$, $\mathcal{F}_{t_{j+1}} \subset \mathcal{F}_{t_k}$, so ΔW_j is $\mathcal{F}_{t_k}$ -measurable.
- H_{t_k} is explicitly $\mathcal{F}_{t_k}$ -measurable by definition.

Because H_{t_j} , ΔW_j , and H_{t_k} are completely determined by the state of the world at time t_k , they are factored out of the inner conditional expectation:

$$\mathbb{E} [\mathbb{E}[H_{t_j} H_{t_k} \Delta W_j \Delta W_k \mid \mathcal{F}_{t_k}]] = \mathbb{E} [H_{t_j} H_{t_k} \Delta W_j \mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}]]$$

As shown previously, $\mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}] = 0$ because Brownian motion increments have independent, zero-mean distributions. Therefore:

$$\mathbb{E} [H_{t_j} H_{t_k} \Delta W_j \cdot 0] = 0$$

Thus, **all cross-product terms vanish identically** due to the martingale property of the forward Brownian increment.

Step 5: Line-by-Line Dissection of Diagonal Terms

We are left with only the diagonal terms of the expansion:

$$\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \sum_{k=0}^{n-1} \mathbb{E} [H_{t_k}^2 (\Delta W_k)^2]$$

Applying the Tower Property again by conditioning each term on $\mathcal{F}_{t_k}$:

$$\mathbb{E} [H_{t_k}^2 (\Delta W_k)^2] = \mathbb{E} [\mathbb{E}[H_{t_k}^2 (\Delta W_k)^2 \mid \mathcal{F}_{t_k}]]$$

Since $H_{t_k}^2$ is $\mathcal{F}_{t_k}$ -measurable, we factor it out:

$$\mathbb{E} [\mathbb{E}[H_{t_k}^2 (\Delta W_k)^2 \mid \mathcal{F}_{t_k}]] = \mathbb{E} [H_{t_k}^2 \mathbb{E}[(\Delta W_k)^2 \mid \mathcal{F}_{t_k}]]$$

Since ΔW_k is independent of $\mathcal{F}_{t_k}$, its conditional variance matches its unconditional variance:

$$\mathbb{E}[(\Delta W_k)^2 \mid \mathcal{F}_{t_k}] = \mathbb{E}[(\Delta W_k)^2] = \text{Var}(W_{t_{k+1}} - W_{t_k}) = t_{k+1} - t_k = \Delta t_k$$

Substituting Δt_k back into the expression yields:

$$\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \sum_{k=0}^{n-1} \mathbb{E}[H_{t_k}^2] \Delta t_k = \mathbb{E} \left[\sum_{k=0}^{n-1} H_{t_k}^2 \Delta t_k \right]$$

Taking the L_2 limit as the partition mesh size $\|\Pi\| \rightarrow 0$, the simple process sum converges to the classical Riemann integral:

$$\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right]$$

This establishes the **Itô Isometry** for all square-integrable adapted processes. Since $\mathbb{E}[I(H)] = 0$, the left-hand side is identically the variance:

$$\text{Var} \left(\int_0^T H_t dW_t \right) = \mathbb{E} \left[\int_0^T H_t^2 dt \right]$$

Step 6: Calculation for a Deterministic Integrand σ_t

If the integrand is a non-random, deterministic function of time σ_t (such as a pre-computed intraday seasonal volatility curve), the expectation operator on the right-hand side of the isometry can be dropped because there is no path dependency:

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] = \int_0^T \sigma_t^2 dt$$

Thus, the variance of the stochastic integral simplifies to:

$$\text{Var} \left(\int_0^T \sigma_t dW_t \right) = \int_0^T \sigma_t^2 dt$$

Because a deterministic integral of Brownian increments is a continuous limit of sums of independent normal random variables, the integral itself is normally distributed:

$$\int_0^T \sigma_t dW_t \sim \mathcal{N} \left(0, \int_0^T \sigma_t^2 dt \right)$$

2. Quantitative Infrastructure & Application of Theory

The following production-grade Python implementation models execution path timing risk within a Transaction Cost Analysis (TCA) framework. It simulates structural intraday volatility curves (the classic U-shaped smile), runs a Monte Carlo engine over thousands of execution trajectories, calculates empirical variances, validates them against the analytical outputs derived via Itô Isometry, and provides an interactive Plotly dashboard.

```
import numpy as np
import pandas as pd
import scipy.integrate as integrate
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# =====
# 1. STRUCTURAL INTRADAY VOLATILITY PATTERN (DETERMINISTIC FUNCTION)
# =====
def get_deterministic_volatility(t: float, base_vol: float = 0.02) -> float:
    """
    Computes deterministic intraday volatility pattern at time t in.
    Replicates the classic empirical 'U-shaped' smile of Cash Equities markets.
    """
    # Diurnal functional form: high at open (t=0), low at midday (t=0.5), high at
    close (t=1)
    smile = 0.05 * (t - 0.5)**2 + base_vol
    return float(smile)

def get_deterministic_volatility_squared(t: float, base_vol: float = 0.02) ->
float:
```

```

    """Returns the squared volatility at time t for variance integrating
    functions."""
    return get_deterministic_volatility(t, base_vol)**2

# =====
# 2. MONTE CARLO COVARIANCE AND INTEGRAL EVALUATOR
# =====
def run_tca_isometry_validation(n_paths: int = 5000, n_steps: int = 1000,
base_vol: float = 0.02) -> tuple:
    """
    Simulates stochastic execution trajectories via Euler-Maruyama.
    Validates empirical path variance directly against analytical Itô Isometry.
    """
    T = 1.0 # Normalized trading horizon (e.g., full market day)
    dt = T / n_steps
    t_space = np.linspace(0, T, n_steps)

    # Pre-calculate deterministic volatility vector across step grid
    vol_vector = np.array([get_deterministic_volatility(t, base_vol) for t in
t_space])

    # Allocate storage for paths
    stochastic_integrals = np.zeros((n_paths, n_steps))

    # Seed noise matrix
    np.random.seed(42)
    dW = np.random.normal(loc=0.0, scale=np.sqrt(dt), size=(n_paths, n_steps))

    # Evaluate paths:  $I_t = \int_0^t \sigma_s dW_s$ 
    for step in range(1, n_steps):
        # Forward Itô increment integration loop
        stochastic_integrals[:, step] = stochastic_integrals[:, step - 1] +
vol_vector[step - 1] * dW[:, step - 1]

    # Final values at time T
    final_realizations = stochastic_integrals[:, -1]

    # Compute Empirical Metrics
    empirical_mean = np.mean(final_realizations)
    empirical_variance = np.var(final_realizations)

    # Compute Analytical Metrics via Numerical Quadrature (Itô Isometry)
    analytical_variance, _ = integrate.quad(get_deterministic_volatility_squared,
0, T, args=(base_vol,))

    return t_space, vol_vector, stochastic_integrals, final_realizations,
empirical_mean, empirical_variance, analytical_variance

# =====
# 3. INTERACTIVE PLOTLY DASHBOARD CANVAS
# =====
def generate_tca_dashboard(t_space, vol_vector, stochastic_integrals,
final_realizations, emp_mean, emp_var, ana_var):
    """

```

```

Renders an interactive high-fidelity visualization platform mapping out
intraday paths, volatility curves, and residual normality diagnostics.
"""
fig = make_subplots(
    rows=2, cols=2,
    subplot_titles=(
        "Deterministic Intraday Volatility Pattern (U-Shape Smile)",
        "Sample Paths of Realized Execution Variance Projections",
        "Empirical vs. Analytical Variance (Itô Isometry Calibration)",
        "Normality Density Curve of Final Stochastic Outcomes"
    ),
    vertical_spacing=0.15,
    horizontal_spacing=0.12
)

# Subplot 1: Volatility Smile Curve
fig.add_trace(go.Scatter(
    x=t_space, y=vol_vector, mode='lines',
    line=dict(color='cyan', width=3), name='Intraday Volatility Curve'
), row=1, col=1)

# Subplot 2: Trajectory Profiles (Plot subset to prevent browser crash)
n_plot_paths = min(60, stochastic_integrals.shape)
for path_idx in range(n_plot_paths):
    fig.add_trace(go.Scatter(
        x=t_space, y=stochastic_integrals[path_idx, :], mode='lines',
        line=dict(width=0.8), opacity=0.4, showlegend=False
    ), row=1, col=2)

# Subplot 3: Variance Identity Matrix Chart
fig.add_trace(go.Bar(
    x=['Empirical Variance (Monte Carlo)', 'Analytical Variance (Itô
Isometry)'],
    y=[emp_var, ana_var],
    marker_color=['lightcoral', 'mediumpurple'],
    text=[f"{emp_var:.6f}", f"{ana_var:.6f}"], textposition='auto'
), row=2, col=1)

# Subplot 4: Outcome Normality Distribution
fig.add_trace(go.Histogram(
    x=final_realizations, nbinsx=45, histnorm='probability density',
    marker_color='teal', opacity=0.75, name='Realized Ends'
), row=2, col=2)

# Superimpose idealized gaussian fit over histogram
x_range = np.linspace(final_realizations.min(), final_realizations.max(), 200)
gaussian_fit = (1.0 / np.sqrt(2.0 * np.pi * ana_var)) * np.exp(- (x_range -
0.0)**2 / (2.0 * ana_var))
fig.add_trace(go.Scatter(
    x=x_range, y=gaussian_fit, mode='lines',
    line=dict(color='gold', width=2, dash='dash'), name='Ideal Normal
Distribution'
), row=2, col=2)

```

```

# Configuration updates
fig.update_layout(
    title_text=f"Nomura Portfolio Execution Analytics Framework – Itô Isometry
Verification Engine",
    title_x=0.5, height=900, width=1300, template="plotly_dark",
    showlegend=False
)

# Set Axis Titles
fig.update_xaxes(title_text="Normalized Trading Session Timeline [0, T]",
row=1, col=1)
fig.update_yaxes(title_text="Instantaneous Asset Volatility  $\sigma(t)$ ", row=1,
col=1)
fig.update_xaxes(title_text="Normalized Trading Session Timeline [0, T]",
row=1, col=2)
fig.update_yaxes(title_text="Stochastic Integral Value  $\int \sigma(s) dW(s)$ ", row=1,
col=2)
fig.update_yaxes(title_text="Calculated Variance Scale", row=2, col=1)
fig.update_xaxes(title_text="Final Realized Trajectory Return Value", row=2,
col=2)
fig.update_yaxes(title_text="Probability Density Frequency", row=2, col=2)

fig.show()

if __name__ == "__main__":
    # Execute quantitative computational validation loop
    t_space, vol_vector, paths, realizations, mean_out, var_emp, var_ana =
run_tca_isometry_validation()

    print("\n" + "="*65)
    print("STOCHASTIC PROCESS SIMULATION MATRIX & ISOMETRY SUMMARY")
    print("="*65)
    print(f"Total Trajectory Test Cohort Size : {paths.shape}")
    print(f"Empirical Simulation Expectation : {mean_out:.6f} (Theoretical
Target: 0.000000)")
    print(f"Empirical Simulation Path Variance : {var_emp:.6f}")
    print(f"Analytical Itô Isometry Integration: {var_ana:.6f}")
    print(f"Precision Identity Mismatch Delta : {abs(var_emp - var_ana):.6f}")
    print("="*65 + "\n")

    # Display Dashboard Canvas
    generate_tca_dashboard(t_space, vol_vector, paths, realizations, mean_out,
var_emp, var_ana)

```

3. Strategic Interview Follow-Ups & Edge-Case Vulnerabilities

Follow-Up 1: The Breakdown of Isometry under Non-Adapted Execution Schedules

Question: "Suppose you are designing an optimal liquidation schedule that uses a signal from an alternate time zone that has already closed. If your execution strategy H_t looks ahead into future

information, such that H_t is not adapted to the historical filtration $\mathcal{F}_t$, does the Itô Isometry still hold? Where does your mathematical proof break down?"

- **The Answer:** The Itô Isometry breaks down completely, and the variance of the integral cannot be simplified to $\mathbb{E} \left[\int H_t^2 dt \right]$.
- **The Mathematical Breakdown:** Look back closely at **Step 4** of the derivation. To vanish the cross-product terms ($j < k$), we evaluated the inner conditional expectation $\mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}]$. Because H_t was adapted, H_{t_k} was fully known at time t_k , allowing us to factor it outside the conditional operator and reduce $\mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}]$ to $\mathbb{E}[\Delta W_k] = 0$.
- If H_t uses future information, it is **non-adapted (anticipating)**. Therefore, H_{t_k} cannot be pulled out of the conditional expectation, as it depends on the future increment ΔW_k . The cross-product terms do not vanish, and the integral must instead be treated using the **Skorokhod integral (Malliavin calculus)** rather than standard Itô calculus.

Follow-Up 2: Physical Application to Timing Risk and Portfolio Downside Limits

Question: "In our Cash Equities Central Risk Book (CRB), we use the expression $\sigma_{\text{execution}}^2 = \frac{1}{T^2} \int_0^T \sigma_t^2 dt$ to represent the timing risk component of Implementation Shortfall. If a client demands a hard guarantee that their tracking error will not exceed a specific variance threshold, how do you adjust your intraday execution schedule to minimize this integral in the presence of our U-shaped volatility smile?"

- **The Answer:** To minimize tracking error variance under a deterministic volatility curve, the execution pace must be optimized dynamically relative to instantaneous volume and volatility.
- Under the Itô Isometry framework, timing risk accumulates fastest during periods where σ_t^2 is elevated—which corresponds to the market open and close. To minimize the variance of the execution shortfall:
 1. We front-load execution volume or increase the participation rate π_t during the market open to clear risk while the local variance parameter is high, reducing the residual variance of the remaining unexecuted shares.
 2. Alternatively, if minimizing market impact is co-equal to minimizing timing risk, we map the execution schedule directly to the **intraday volume smile**. By trading a constant proportion of expected market volume, we ensure that the time parameters are rescaled from chronological time dt to **volume time** dv . In volume time, the asset's volatility becomes approximately constant (homoscedastic), stabilizing the tracking error and insulating the CRB capital from sudden intraday volatility regimes.